When Fusion Meets Super-Resolution: Implicit Edge Calibration for Higher Resolution Multispectral Image Reconstruction

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***Abstract*— Multispectral image (MSI) reconstruction via remote sensing image fusion (RSIF), involving the fusion of MSIs with panchromatic (PAN) images, has attracted considerable attention. However, the main limitation of RSIF lies in the fact that the resolution of the fused image is limited by the PAN image. Although single image super-resolution (SISR) methods have the potential to improve image resolution, the larger scale factors often result in the loss of edge sharpness. To tackle these challenges, this article proposes a novel end-to-end framework for MSI reconstruction, termed FSRNet, aimed at generating higher resolution MSI while preserving the fine edge structural details. The FSRNet seamlessly integrates RSIF and SISR into a unified framework, thereby mitigating the cumulative errors that typically arise in two-stage cascaded methods. Moreover, an innovative implicit edge calibration (IEC) scheme is proposed that leverages gradient prior knowledge implicitly to calibrate and sharpen the MSI structure effectively. Notably, IEC is exclusively utilized during the training phase and is omitted during testing. Thus, it does not increase the complexity of the network in the actual application (i.e., testing phase). Additionally, we introduce the progress feature distillation module to improve feature representation through the feature distillation structure, facilitating the learning of more distinctive hierarchical features. Experimental results on the QuickBird, Gaofen-2, and WorldView-3 datasets demonstrate that the proposed method exhibits advantages compared to other state-of-the-art SISR and two-stage methods in terms of both quantitative and qualitative comparisons.**

***Index Terms*— Implicit edge calibration (IEC), progressive feature distillation, remote sensing image fusion (RSIF), single image super-resolution (SISR).**

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I. Introduction

MULTISPECTRAL image (MSI) plays a crucial role in remote sensing applications by providing valuable insights into Earth's surface features across various wavelengths [1]. However, multispectral sensors prioritize spectral information over spatial resolution, resulting in images with lower spatial clarity. Therefore, the reconstruction of high-resolution (HR) MSIs from satellite or aerial observations is significant in remote sensing image processing. It aims to facilitate the extraction of detailed information about land cover, vegetation health, urban development, and other geospatial phenomena [2]. Obtaining a higher resolution MSI allows researchers to achieve a more comprehensive understanding of complex environmental process. The higher the resolution of reconstructed MSIs, the more beneficial it is for subsequent downstream tasks such as feature extraction [3], change detection [4], and semantic segmentation [5], leading to insightful conclusions. Researchers have therefore explored innovative methods continuously to improve the spatial resolution of reconstructed MSIs. Currently, the methods for acquiring HR MSI can be categorized into remote sensing image fusion (RSIF) [6] and single image super-resolution (SISR) [7].

RSIF aims to fuse HR panchromatic (PAN) images with low-resolution (LR) MSIs to generate spatially enhanced MSIs. Some traditional RSIF algorithms are proposed to advance the field of remote sensing applications [8]. Palsson et al. [9] proposed an RSIF algorithm based on total variation, which is based on the assumption that the PAN image is a linear combination of the bands of the pansharpened image. And it is assumed that by decimating the pansharpened image, the original MSI can be obtained. He et al. [10] proposed a variational model based on spatial and spectral sparsity priors for the RSIF. With the development of deep learning [11], [12], [13], convolutional neural networks (CNNs) are widely adopted in RSIF. For example, Yuan et al. [14] propose the multiscale and multidepth CNN for RSIF by introducing multiscale feature extraction and residual learning into the basic CNN architecture. To fully exploit the complementary modality of the two images, a novel and efficient asymmetric bidirectional fusion network [15] is proposed.

SISR aims to enhance the spatial resolution of images by reconstructing HR images directly from LR counterparts, thereby providing high-quality images for downstream

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high-level tasks such as change detection and classification [16], [17], [18]. Traditional methods for obtaining HR images relied heavily on interpolation-based upsampling techniques, often resulting in a significant loss of high-frequency information [19]. In contrast, deep learning-based approaches leverage their powerful learning capabilities to achieve superior results by effectively mapping LR images to HR images. One of the pioneering works in this field was done by Dong et al. [20], who designed a network with three convolutional layers specifically for image super-resolution. Building on this foundation, Lei et al. [21] introduced a local–global network tailored for remote sensing image super-resolution, which simultaneously extracts both global and local features to improve image quality. Furthermore, a stable and interpretable network is proposed by Guo et al. [22] to enhance the interpretability and efficiency of the super-resolution process. This interpretable network improves the understanding of how the model works and offers significant efficiency advantages.

However, with the emergence of several excellent RSIF and SISR methods, they are often unsatisfactory in reconstructing MSI with higher spatial resolution. It remains a difficult task for the following reasons.

1) How can we reconstruct MSIs beyond the spatial resolution limits of HR auxiliary images like PAN images? Most RSIF methods train a model at the given scale of LR MSI and PAN image, but practical applications often demand higher resolutions. Therefore, a method is needed that can break through the resolution constraints of PAN images to generate higher resolution MSIs.
2) How to exploit the edge prior knowledge rather than treating the MSI reconstruction process as a simple regression problem in deep learning-based methods? Several existing methods explicitly utilize edge information by constructing an additional edge auxiliary network as part of the main network. This, however, makes the network more complex and difficult to optimize.
3) How to address the problem that traditional networks may suffer from poor feature representation, leading to a diminished ability to represent complex remote sensing data? This limitation is due to the serial and parallel convolutional architectures failing to progressively extract hierarchical features, resulting in a loss of crucial details.

To address these problems, in this article, we propose a novel end-to-end MSI reconstruction framework, named FSRNet. It seamlessly integrates RSIF and SISR into a unified framework while subtly leveraging the strengths of both to reconstruct higher resolution MSI. By optimizing an overall objective function, this framework can simultaneously consider the RSIF and SISR, improving image clarity while reducing cumulative errors. To fully leverage the edge prior knowledge, an implicit edge calibration (IEC) scheme is proposed to reconstruct MSI with high accuracy and fine details. Especially, the "implicit" is used because IEC is employed only in the training phase and is omitted in testing. This means that it does not add to the complexity of the network during the testing phase, i.e., in the actual application. Furthermore, to alleviate the problem of poor feature representation in traditional serial and parallel convolutional networks, the progressive feature distillation module (PFDM) is introduced to distill the essential features of the intermediate layers and to refine the features in a progressive manner. In summary, the main contributions of this work are threefold.

1) We propose an innovative end-to-end framework that improves the spatial resolution of MSIs without being limited by the spatial resolution of PAN images. This means that the proposed method requires only one well-trained model to generate MSIs with higher spatial resolution while reconstructing better details.
2) Besides the spatial domain, we propose an IEC scheme to enhance the structural details of MSIs implicitly by exploiting the gradient domain, effectively calibrating edge structure features of the main network during the training. In the testing phase, it is omitted and therefore does not increase the network complexity in practice.
3) We introduce a PFDM that progressively refines and deepens the feature extraction process through a progressive distillation connection. This enables the extraction and refinement of features from various levels, facilitating the capture of complex structures within the MSIs.

The remaining work is divided into four sections. Section II describes the related work on RSIF, SISR, and gradient-relevant methods. In Section III, the proposed FSRNet is described in detail. The experimental results are presented in Section IV. The conclusion is given in Section V.

II. Related Work

Some of the more relevant representatives, including RSIF, SISR, and gradient-relevant methods, are briefly discussed in this section.

A. RSIF Methods

Research in RSIF has made substantial progress, leading to the proposal of numerous methodologies. Broadly, RSIF approaches are categorized into component substitution, multiresolution analysis, variational optimization, and deep-learning-based methods.

The component substitution method involves projecting the LR MSI into a transformed domain, replacing its spatial components with corresponding details from the PAN image. Carper et al. [23] demonstrated the effectiveness of a weighted fusion of PAN and near-infrared data for enhancing image quality. Multiresolution analysis decomposes the LR MSI into multiple resolutions to extract spatial information, with Do and Vetterli [24] introducing a 2-D transform capturing crucial geometric structures important for visual information. Variational optimization methods integrate prior information to construct regularization terms, enhancing the spectral fidelity of MSIs. Fang et al. [25] pioneered regularization terms based on sparse priors to improve the quality of MSIs. Besides, Yin et al. [26] achieved better image fusion using a separated super-resolution based on sparse representation. However, these methods are constrained in flexibility and efficacy, particularly in spatial resolution enhancement and spectral fidelity.

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With the rapid development of deep learning, the application of deep learning techniques to RSIF has gained significant attention [27]. Leveraging the autoencoder paradigm, Huang et al. [28] earlier applied deep learning to RSIF. Subsequent advances saw the introduction of CNN-based RSIF networks, with Masi et al. [29] pioneering a three-layer convolution method. Techniques like residual learning have mitigated gradient issues, as seen in the work of Wei et al. [30]. To extract multiscale features, Chi et al. [31] introduced a multiscale pyramid network and achieved excellent results. Deng et al. [32] innovatively integrated CNN methods with traditional physical concepts to propose a novel RSIF network (FusNet), enhancing model interpretability. Most CNN architectures adopt standard convolutional operations, but dynamic convolution is impressive in high-level tasks yet rarely applied in RSIF. Hu et al. [33] proposed a multiscale RSIF method based on dynamic CNN and obtained better results. Jin et al. [34] introduced a local adaptive kernel generation and global bias mechanism to enhance feature extraction capability. Inspired by back projection principles, Zhang et al. [35] developed the spatial spectral dual backprojection engineered network (S2DBPN), leveraging back projection in spatial and spectral domains to fuse LR MSI and HR PAN image. These advanced methods are excellent for RSIF tasks, albeit facing limitations in obtaining higher-resolution MSIs.

B. SISR Methods

SISR involves multiple mappings from LR to HR space, leading to various potential solution spaces for any given LR input [36]. These methods are typically classified as interpolation-based [19], reconstruction-based [37], and deep learning-based methods [38]. Interpolation-based methods, such as bicubic interpolation, though simple and efficient, often fail to preserve high-frequency image details. Reconstruction-based methods incorporate prior image knowledge as constraints during the SISR process. For instance, Dai et al. [39] introduced an adaptive super-resolution method utilizing edge smoothing based on the SoftCuts metric, albeit with slow convergence and high computational costs. Deep learning-based methods leverage the powerful learning capabilities of neural networks and vast LR and HR image datasets. Early work by Dong et al. [20] pioneered deep learning-based SISR with a three-layer CNN framework. To optimize memory usage and computational efficiency, Wang et al. [40] presented a lightweight feature-enhanced network, FeNet, specifically tailored for remote sensing image super-resolution. Recent advancements have significantly shifted toward Transformer-based and diffusion-based approaches. Liang et al. [41] introduced a Swin Transformer architecture (i.e., SwinIR) for image restoration that efficiently captures long-range dependencies, enhancing super-resolution performance. To activate more input pixels for better reconstruction, a novel hybrid attention Transformer (HAT) is proposed [42]. Xiao et al. [43] then proposed a top-k token selective Transformer (TTST), which improves self-attention efficiency by adaptively filtering irrelevant tokens and obtains better super-resolution results. They employed diffusion models (i.e., EDiffSR) for SISR [44], offering a novel approach to generate high-fidelity details in super-resolved images. Despite these advancements, previous SISR methods often struggle to effectively reconstruct HR images with large scale factors, highlighting the need for continued innovation in this field.

C. Gradient-Relevant Methods

Gradients play a crucial role in capturing structural information [45], [46], thus they are applied in a wide range of tasks. To meet the requirements of fine-grained image segmentation, Zhang et al. [47] proposed a collaborative boundary-aware context encoding framework in which the extraction of boundary information is crucial. Ye et al. [48] proposed a pseudo-label correction method for domain-adaptive medical image segmentation, highlighting the crucial role of edge information in achieving accurate segmentation results. Zhou et al. [49] proposed improving the boundary perception capability of the network by learning the distance relationship between foreground–background regions and boundaries. This idea has also been introduced into the image restoration tasks. Jiang et al. [50] introduced an edge-enhanced GAN to improve super-resolution in remote sensing images, emphasizing the significance of edge information. Yan et al. [51] proposed a super-resolution method based on the sharpness of gradient contours extracted from a gradient description model. While this method models HR edge-related parameters based on observed HR correlations in LR images, the process is intricate and lacks flexibility. Recognizing the value of integrating image edge prior knowledge, Fang et al. [52] advocated for the incorporation of such knowledge into a super-resolution model to enhance image resolution, leading to the development of the edge-assisted network. Ma et al. [45] achieved improved results by augmenting a gradient branch within the main network branch to impart gradient knowledge. Barman et al. [53] explored edge-preserving techniques using sparse representations, further demonstrating the importance of edge information in remote sensing image super-resolution. Liu et al. [54] designed gradient spatial attention for hyperspectral image fusion and achieved superior results. Subsequently, Zheng et al. [55] proposed the edge-conditioned feature transform network for multispectral and hyperspectral image fusion, demonstrating superior performance through the adaptive transformation of the gradient prior. This method enhances spatial detail reconstruction by incorporating an auxiliary gradient guidance network directly. However, the explicit nature of these methods increases network complexity while potentially diminishing practical applicability.

In this article, we propose an end-to-end framework that seamlessly integrates RSIF and SISR methods to reconstruct MSIs, aiming at reconstructing higher resolution MSIs beyond the resolution limitations of auxiliary images. Meanwhile, to calibrate and strengthen the edge structure of MSIs, we propose an IEC scheme, which implicitly transfers the edge prior knowledge from the gradient domain to the spatial domain during training. During the testing phase, the gradient-guided

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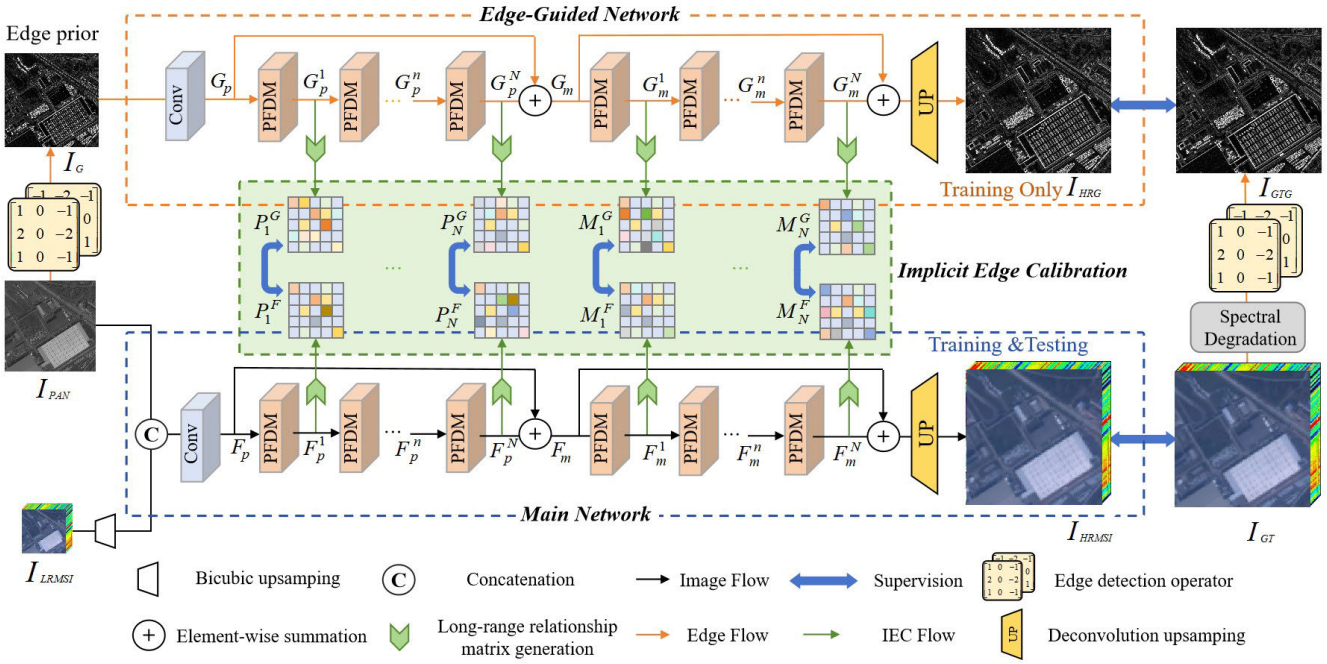


Fig. 1. Pipeline of FSRNet. The overall network consists of two branch networks, i.e., edge-guided network and main network for gradient prior knowledge acquisition and MSI reconstruction. The IEC branch calibrates and sharpens the structural features of the main networks.

network is omitted and therefore does not add to the complexity of the network. In the network, the PFDM is introduced to enhance the feature representation by refining distinctive hierarchical features.

III. Proposed Method

A. Overall Framework

The workflow of the FSRNet is depicted in Fig. 1, which comprises an edge-guided network and a main network, both constructed through the stacking of PFDMs. Moreover, the IEC is included to implicitly transfer gradient prior knowledge between the two networks, thereby calibrating and enhancing the reconstruction of edge structures within the MSIs. The $I_{\text{HRMSI}} \in \mathbf{R}^{C\times H\times W}$ and $I_{\text{GT}} \in \mathbf{R}^{C\times H\times W}$ represent the output reconstructed MSI and the ground truth (GT), where C, H, and W represent spectral bands, height, and width, respectively. The $I_{\text{PAN}} \in \mathbf{R}^{1\times(H/4)\times(W/4)}$ and $I_{\text{LRMSI}} \in \mathbf{R}^{C\times(H/16)\times(W/16)}$ denote the input PAN image and LR MSI. Our goal is to reconstruct higher resolution MSIs (i.e., I_{HRMSI}) by the end-to-end framework. The generation of I_{HRMSI} from the proposed FSRNet can be reformulated as

$$I_{\text{HRMSI}} = f_\theta(I_{\text{LRMSI}} \uparrow, I_{\text{PAN}} | I_G) \tag{1}$$

where $\uparrow$ represents the bicubic upsampling operation, $f_\theta(\cdot)$ stands for the end-to-end framework parametrized by θ, and I_G represents the edge prior knowledge that implicitly calibrates the main network to accurately reconstruct the higher resolution MSI.

1) Main Network: First, the I_{LRMSI} is upscaled to obtain the same spatial size as I_{PAN}. For the upscaled I_{LRMSI} and I_{PAN}, a convolutional layer is utilized to obtain initial features

$$I_{\text{PAN}} = f_{\text{conv3}\times3}([I_{\text{LRMSI}} \uparrow, I_{\text{PAN}}]) \tag{2}$$

where $f_{\text{conv3}\times3}$ represents 3×3 convolution operation.

Then, after obtaining these shallow features, they are fed into multiple PFDMs and accompanied by residual connection to extract mid-level features F_m. This learning process can be represented as

$$F_m = f_{\text{PFDM}_N}(\cdots(f_{\text{PFDM}_1}(F_p))) + F_p \tag{3}$$

where f_{PFDM} represents progressive feature distillation module.

Finally, the mid-level features are fed into multiple PFDMs, residual connection, and deconvolutional upsampling to reconstruct the final I_{HRMSI}

$$I_{\text{HRMSI}} = f_{\text{up}}(f_{\text{PFDM}_N}(\cdots(f_{\text{PFDM}_1}(F_m))) + F_m) \tag{4}$$

where f_{up} represents deconvolutional layer.

2) Edge-Guided Network: In addition, we propose to leverage the gradient prior knowledge from the edge-guided network to strengthen the reconstruction capability of the structural details of the main network. First, the edge detection operator is utilized to transmit the PAN image into the gradient domain to obtain an edge prior. Specifically, given the PAN image, the process of acquiring the edge prior is

$$I_G = \sqrt{(I_{\text{PAN}} * G_x)^2 + (I_{\text{PAN}} * G_y)^2} \tag{5}$$

where I_G is the generated edge prior. The G_x and G_y are the horizontal and vertical convolution kernels of the Sobel operator [56].

Then, similar to the main network, the edge prior is fed into convolutions for initial feature extraction. Next, multiple PFDMs are used to obtain edge features at different stages. These can further enrich the edge feature representation ability of the edge-guided network. Finally, I_{HRG} is obtained by

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upsampling the edge feature map to the same spatial size as the HR MSI via a deconvolutional layer.

B. Implicit Edge Calibration

As acknowledged, the reconstruction of HR images directly from LR images often faces the problem of blurred structural edges. To ensure the accuracy of MSI reconstruction, several works consider strengthening the learning of edge representations in the main network [45], [55]. The gradient prior provides important information about image edges and variations, contributing to improved image sharpness. Motivated by this insight, we propose an IEC scheme that incorporates the edge prior knowledge as an additional supervision signal to facilitate the ability of the main network to capture image structures and finer details effectively. Unlike prior works that require additional auxiliary networks to directly inject edge features, the proposed IEC scheme is only utilized during the training phase and not during testing, ensuring minimal impact on network complexity.

Fig. 1 illustrates the specifics of the proposed IEC. The LR MSI and PAN image are fed into the main network, while the PAN image is also fed into the edge-guided network. The edge-guided network can extract the edge prior of the PAN image and obtain the edge features of different layers through the deep network. Notably, both networks adopt a consistent architecture but serve distinct functions within the framework. To efficiently transfer the edge prior information from the edge-guided network to the main network, the feature maps of the main network are forced to mimic the feature map from the edge-guided network by minimizing the long-range relationship matrix. The key of IEC is to design a suitable simulated loss function that can successfully propagate valuable information (i.e., edge prior knowledge) to guide the training process of the main network.

Given a batch of feature maps $G_p^n \in \mathbf{R}^{b\times C\times H\times W}$ and $F_p^n \in \mathbf{R}^{b\times C\times H\times W}$ from edge-guided network and main network, respectively. The b stands for batch size. To implicitly exploit the edge prior on the global space, a long-range relationship matrix is required to be calculated. Specifically, we reshape G_p^n and F_p^n into $b \times C \times HW$ firstly. Then, the feature map is multiplied with the transpose of itself by normalization. For instance, the P_n^F is formulated as

$$P_n^F = \left(\frac{F_p^n}{\left\|F_p^n\right\|_2^2}\right) \cdot \left(\frac{F_p^n}{\left\|F_p^n\right\|_2^2}\right)^T \tag{6}$$

where the size of the generated long-range relationship matrix P_n^F is $b \times HW \times HW$. Each element of the long-range relationship matrix represents the spatial dependency between two pixels. The G_n^F, M_n^G, and M_n^F are obtained in the same way. The main network is encouraged to generate a similar long-range relationship matrix with the edge-guided network to strengthen the learning of edge structural features, and the loss constraint of IEC can be formulated as

$$L_{\text{IEC}} = \frac{1}{|P|}\sum_{n=1}^{N}\left\|P_n^F - P_n^G\right\|_1 + \frac{1}{|M|}\sum_{n=1}^{N}\left\|M_n^F - M_n^G\right\|_1 \tag{7}$$

where P_n^F and M_n^F represent the long-range relationship matrix from the main network, P_n^G and M_n^G represent the long-range relationship matrix from the edge-guided network. The $|P|$ and $|M|$ denote the number of elements in the long-range relationship matrix. These two loss functions represent different stages of edge guidance that jointly calibrate the main network to reconstruct better structural details of MSIs.

C. Progressive Feature Distillation Module

Inspired by the efficient separable distillation block [57], we introduce the PFDM, as shown in Fig. 2(a). The PFDM progressively refines the feature extraction process through step-by-step distillation structures. This approach facilitates the extraction and refinement of features at different levels, enabling the capture of complex structures within MSI data. Specifically, the PFDM includes progressive feature distillation, feature aggregation, and feature enhancement. As part of this, the progressive feature distillation can be formulated as

$$\begin{aligned} F_{c1}, F_{s1} &= f_{\text{conv1}\times1}(F_{\text{in}}), f_{\text{SCU}}(F_{\text{in}}) \\ F_{c2}, F_{s2} &= f_{\text{conv1}\times1}(F_{s1}), f_{\text{SCU}}(F_{s1}) \\ F_{c3}, F_{s3} &= f_{\text{conv1}\times1}(F_{s2}), f_{\text{SCU}}(F_{s2}) \\ F_{c4} &= f_{\text{SCU}}(F_{s3}) \end{aligned} \tag{8}$$

where $f_{\text{conv1}\times1}$ represents the distillation operation that generate distilled features by 1×1 convolution. The f_{SCU} represents separable convolution unit (SCU), as shown in Fig. 2(b). This can progressively refine the features, consisting of a pair of separation convolution (i.e., 1×3 and 3×1 convolutions) and a GELU activation function. In particular, the last f_{SCU} consists of only one pair of separated convolutions and reduces the number of channels of feature maps from C to C/2. The next step is to aggregate the distilled features from the different stages and then condense them by a 1×1 convolution as follows:

$$F_c = f_{\text{conv1}\times1}([F_{c1}, F_{c2}, F_{c3}, F_{c4}]). \tag{9}$$

In the feature enhancement phase, to elevate the representation capacity of the model, we introduce the combined attention module (CAB). The CAB comprises spatial attention and channel attention components operating in parallel, as illustrated in Fig. 2(c). The spatial attention mechanism focuses on the spatial relationships among pixels, enhancing the details and textures of HR images. The channel attention mechanism improves the accuracy of the image reconstruction by focusing on different feature channels of the feature maps and learning their interdependencies. Specially, spatial attention enhances spatial features by calculating average and maximum values across the feature map at each position and integrating them to determine positional weights. Meanwhile, channel attention first generates channel features through global average pooling and max pooling, then generates channel descriptors by capturing inter-channel correlation through the fully connected neural network, and finally generates channel attention weight by the sigmoid function. These weights, derived from both attention mechanisms, are then used to modulate F_c, significantly enriching the model's representation

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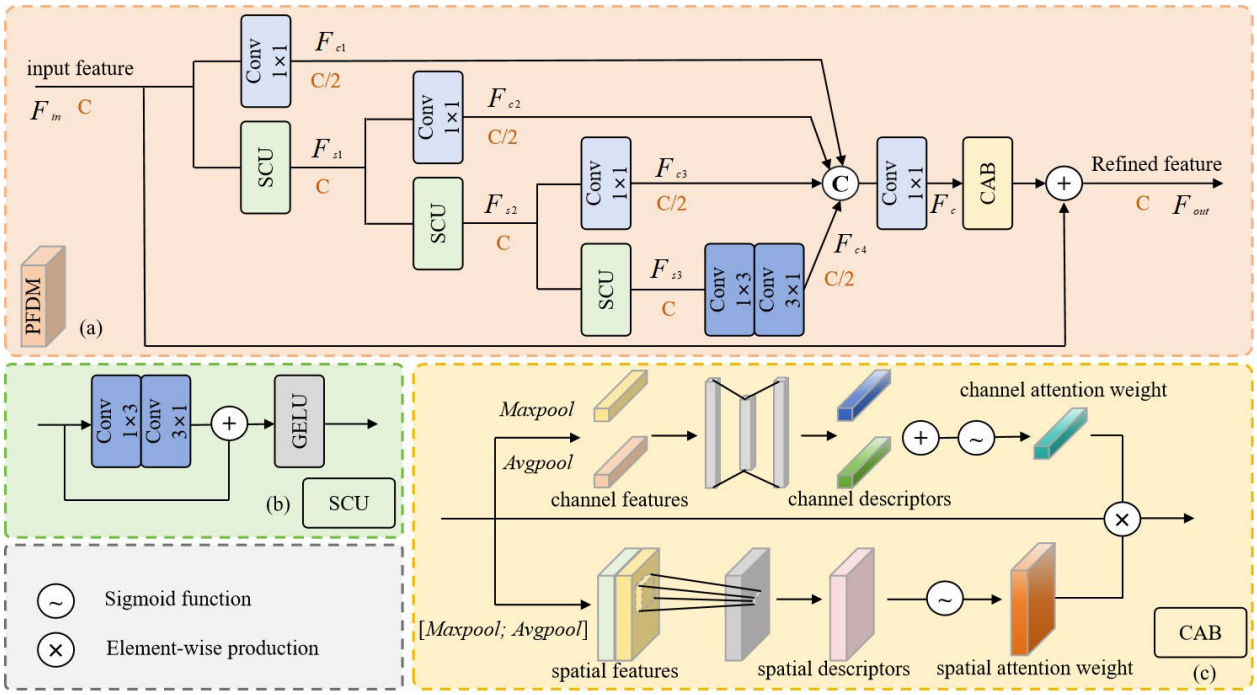


Fig. 2. Illustrations of (a) PFDM, (b) SCU, and (c) CAB.

capabilities. Ultimately, the entire PFDM process can be represented as follows:

$$F_{\text{out}} = f_{\text{CAB}}(F_c) + F_{\text{in}} \tag{10}$$

where f_{CAB} represents the combined attention module. The final F_{out} obtained is the refined feature through progressive feature distillation.

D. Overall Loss Function

Loss functions are pivotal components in deep learning-based supervised tasks. To implicitly calibrate the structural features from the main network, we propose the IEC scheme in the FSRNet, which successfully propagates edge prior knowledge in the global space by minimizing the loss values to guide the reconstruction process of the main network. The total loss function primarily encompasses constraints related to the GT, edge gradient map, and IEC. Specifically, for the supervision of GTs, we employ the L_1 loss to evaluate the accuracy of the reconstructed results. Because the L_1 loss does not over-penalize large errors compared to the L_2 loss and shows better convergence performance [58]. We adopt L_1 loss to measure the reconstruction accuracy. This process can be expressed as

$$L_R = \|I_{\text{HRMSI}} - I_{\text{GT}}\|_1. \tag{11}$$

The gradient maps can describe the edge information of the image well. Therefore, to recover finer image details, the gradient loss is developed as

$$L_G = \|I_{\text{HRG}} - I_{\text{GTG}}\|_1 \tag{12}$$

where I_{HRG} represents the reconstructed HR gradient map and I_{GTG} represents the GT gradient map generated by the spectral degradation of the GTs followed by an edge detection operator. The overall loss function is formulated as

$$L_{\text{total}} = L_G + L_R + \alpha \cos\frac{m\pi}{2M} L_{\text{IEC}} \tag{13}$$

where M and m represent the total training epochs and the current epoch, respectively. α is trade-off coefficient. As the training process progresses, the calibration of the IEC should be gradually reduced, thus transferring the attention of the training to the main network.

IV. Experiments

The purpose of this section is to demonstrate the effectiveness of the proposed method. First, three datasets and implementation details are presented. Second, the comparing methods and evaluation metrics are described. Then, ablation experiments are performed. These experiments demonstrate the effectiveness of the proposed method. Finally, the advantages of the proposed method are verified by comparing the proposed end-to-end framework with SISR and two-stage (i.e., RSIF + SISR) methods on three datasets. This includes experiments with simulated and real data.

A. Datasets and Implementation Details

1) Datasets: We employ three widely used datasets in our experiments, including the QuickBird dataset, the Gaofen-2 dataset, and the WorldView-3 datasets [32], [34]. The QuickBird dataset consists of PAN images with 1 band and MSIs with 4 bands. The spatial sampling intervals are 0.61 and 2.44 m, with a radiometric resolution of 11 bits. It was acquired over Indianapolis. The GaoFen-2 dataset comprises PAN images with 1 band and MSIs with 4 bands. Spatial sampling intervals are 1 and 4 m, and radiometric resolution

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is 10 bits. It was acquired in Guangzhou. The WorldView-3 dataset comprises PAN images with 1 band and MSIs with 8 bands. It includes images from WorldView-3 Rio and WorldView-3 Tripoli, offering spatial sampling intervals of 0.3 and 1.2 m, respectively. Radiometric resolution is 11 bits.

Specifically, each dataset is divided into non-overlapping training and testing regions. To generate training data, the source images are first simulated degraded using Gaussian filters that match the modulation transfer function of the sensor, and then downsampled with a scale factor of 16. The original MSI is considered as the GT, following the Wald's protocol [59]. The experimental testing data includes simulated and real data. The former is acquired in a consistent manner with the training data, using the original MSI as the GT; the latter is the MSI and the PAN image at the original scale and without the GT. Then, after cropping, three training datasets and the corresponding testing datasets are as follows. The QuickBird training dataset consists of 17 139 image pairs. The QuickBird testing dataset consists of simulated data (i.e., PAN image: $128\times128\times1$, MSI: $32\times32\times4$, GT: $512\times512\times4$) and real data (i.e., PAN image: $512\times512\times1$, MSI: $128\times128\times4$). The GaoFen-2 training dataset consists of 19 805 image pairs. The GaoFen-2 testing dataset consists of simulated data (i.e., PAN image: $128\times128\times1$, MSI: $32\times32\times4$, GT: $512\times512\times4$) and real data (i.e., PAN image: $512\times512\times1$, MSI: $128\times128\times4$). The WorldView-3 training dataset consists of 9714 image pairs. The WorldView-3 testing dataset consists of simulated data (i.e., PAN image: $128\times128\times1$, MSI: $32\times32\times8$, GT: $512\times512\times8$) and real data (i.e., PAN image: $512\times512\times1$, MSI: $128\times128\times8$).

2) Implementation Details: In the proposed FSRNet, the Sobel operator, which utilizes 3×3 convolutional kernels, is employed for edge detection to enhance the extraction of gradient information in both horizontal and vertical directions. The ten PFDMs are set up in both the main and the edge guidance networks, respectively, and the number of channels in each PFDM is 64. There are four SCUs and one CAB in the PFDM. The SCU contains 1×3 and 3×1 convolutions, residual connection, and GELU activation functions. The CAB contains channel attention and spatial attention in parallel. Max pooling and average pooling are applied separately along the channel and spatial dimensions to generate attention weights. The α is set to 0.5. The optimization of the network is performed using ADAM [60] optimizer. The initial learning rate is set to 0.0001, halved every 50 epochs, and terminated after 500 epochs. The batch size is set to 32. The network is implemented on an NVIDIA RTX 4090 GPU using the PyTorch framework.

B. Comparing Methods and Evaluation Metrics

1) Comparing Methods: To evaluate the superiority of FSRNet, the proposed method is compared with several state-of-the-art methods including SISR and two-stage methods, as these are standard methods in HR MSI reconstruction. Comparing with SISR methods provides a benchmark for evaluating the advantages of our framework without auxiliary images, while comparison with two-stage methods effectively demonstrates the advantages of the proposed method when including spatial information images (such as the PAN images). Two-stage methods are selected as benchmarks, first image fusion followed by SISR. To ensure a fair comparison, the best performing SISR method is chosen for the second stage. The baseline SISR methods include Bicubic [61], FeNet [40], contextual transformation network (CTNet) [62], hybrid-scale self-similarity exploitation network (HSENet) [63], HAUNet [64], and TTST [43] with a scale factor of 16 for direct capture mapping from LR MSI to HR MSI. The baseline two-stage methods include wavelet-based methods (AWLP) [65] + TTST [43], haze correction method (BRHM) [66] + TTST [43], FusNet [32] + TTST [43], double U-shape network (U2Net) [67] + TTST [43], S2DBPN [35] + TTST [43], and AWFLN [68] + TTST [43]. The results of all comparison methods are based on training with 16 × GTs.

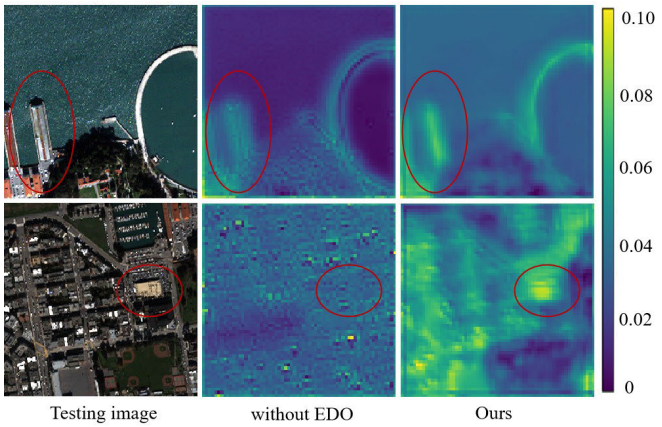


Fig. 3. Visualization of intermediate feature maps F_m.

2) Evaluation Metrics: In the experiments on simulated data, five image quality metrics are utilized to assess the effectiveness of the proposed method, including universal image quality metric (Q) [69], spectral angle mapping (SAM) [70], relative global dimensionless error synthesis (ERGAS) [71], spatial correlation coefficient (SCC) [72], and N-band image fusion quality evaluation metric (Q_N) [73]. These metrics are applied to experiments using simulated data and serve different purposes. The Q aims for values closer to 1 indicating higher quality. The SAM strives for values closer to 0, signifying increased spectral similarity. The ERGAS aims for values closer to 0, indicating a lower error. Additionally, the SCC measures the spatial consistency, with values nearing 1 denoting a strong correlation. The Q_N targets values close to 1 for optimal fusion performance.

In the experiments on real data, no-reference image quality evaluation metrics are adopted, including natural image quality evaluator (NIQE) [74] and perception-based image quality evaluator (PIQE) [74]. In addition, the spectral distortion metric D_λ, the spatial distortion metric D_s, and quality without reference (QNR) value are employed [75].

C. Ablation Experiments

To validate the effectiveness of FSRNet, a series of ablation experiments are conducted on the QuickBird dataset.

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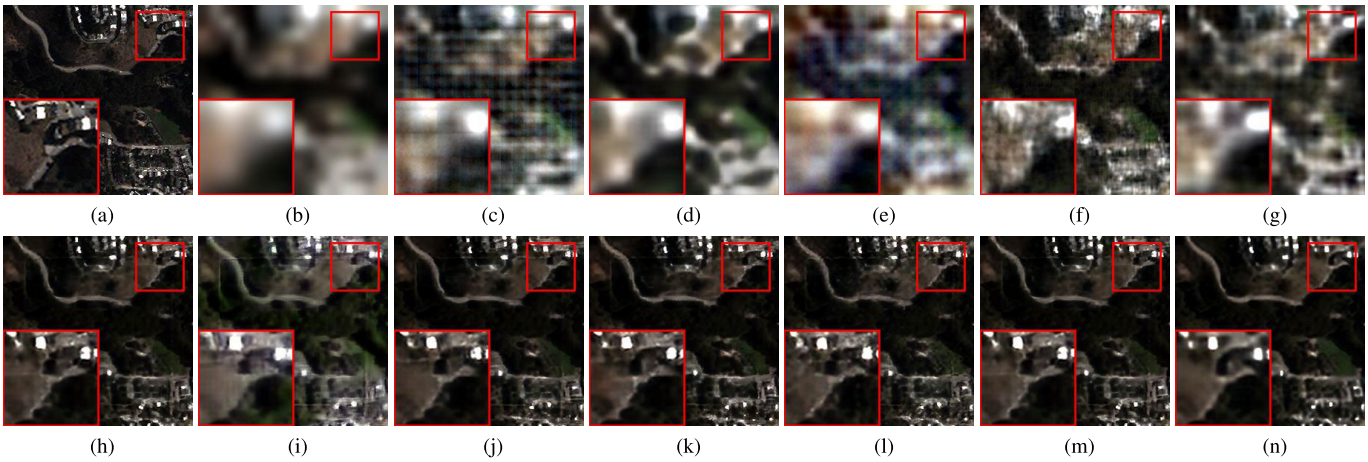


Fig. 4. Reconstruction results of simulation experiments on the QuickBird dataset. (a) GT. (b) Bicubic. (c) FeNet. (d) CTNet. (e) HSENet. (f) HAUNet. (g) TTST. (h) AWLP + TTST. (i) BRHM + TTST. (j) FusNet + TTST. (k) U2Net + TTST. (l) S2DBPN + TTST. (m) AWFLN + TTST. (n) Ours.

TABLE I

STUDY ON THE NUMBER OF PFDMS ON THE QUICKBIRD DATASET. THE BOLD INDICATES THE BEST VALUE

Numbers	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
1	0.6781	8.9199	2.6222	0.5946	0.6775
2	0.6821	8.9063	2.6123	0.5987	0.6827
3	0.6852	8.9121	2.6068	0.6028	0.6859
4	0.6890	8.7734	2.5907	0.6074	0.6890
Ours (5)	**0.6952**	**8.6969**	**2.5728**	**0.6194**	**0.6955**
6	0.6828	8.8523	2.6245	0.5945	0.6822

TABLE II

STUDY ON THE IEC ON THE QUICKBIRD DATASET. THE BOLD INDICATES THE BEST VALUE

	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
w/o EDO	0.6871	8.7635	2.6335	0.5987	0.6869
w/o IEC	0.6874	8.7312	2.6480	0.5894	0.6884
w/o FGL	0.6939	8.7207	2.5742	0.6141	0.6942
w/o WD	0.6822	8.8994	2.6114	0.6028	0.6820
w/o LRMG	0.6946	8.7050	2.5793	0.6150	0.6953
Ours	**0.6952**	**8.6969**	**2.5728**	**0.6194**	**0.6955**

TABLE III

STUDY ON THE PFDM ON THE QUICKBIRD DATASETS. THE BOLD INDICATES THE BEST VALUE

	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
w SC	0.6895	8.7913	2.5905	0.6147	0.6900
w PC	0.6930	8.7416	2.5796	0.6131	0.6934
w 3×3 Conv	0.6870	8.7287	2.6344	0.6055	0.6885
w/o CAB	0.6933	8.7220	2.5733	0.6147	0.6934
w/o HFE	0.6792	8.9021	2.6150	0.6016	0.6796
Ours	**0.6952**	**8.6969**	**2.5728**	**0.6194**	**0.6955**

1) Number of PFDMs: The number of PFDM is varied from 1 to 6 to analyze the impact of module quantity on the framework. Table I presents the results for five metrics, showing that higher values of N correspond to better performance. This is because a deeper network offers superior model capabilities. However, when N is set to 6, the evaluation metrics degrade. Deeper network is harder to train, which explains this phenomenon. In subsequent experiments, N is fixed at 5 in the FSRNet.

2) Performance of IEC: To assess the benefits of the IEC scheme, relevant ablation experiments are designed, and the results are displayed in Table II. The results indicate that the value of Q significantly decreases after removing the edge detection operation (EDO), as EDO is more effective at extracting spatial edge details. Additionally, the visualizations of the intermediate feature maps F_m before and after removing EDO are shown in Fig. 3. As illustrated by the red circles, incorporating EDO increases the network's attention to edge regions, which helps with image reconstruction. Removing the IEC leads to an overall decrease in metrics. The IEC is able to excellently calibrate the training of the main network and plays a crucial role in sharpening the edge structure of MSIs. Removing the final gradient loss (FGL) also leads to performance degradation, which highlights its role in reconstructing high quality MSI. In the training of FSRNet, the drop in performance after weight decay (WD) removal demonstrates the role of WD in improving the effectiveness of MSI reconstruction. Moreover, the removal of long-range relationship matrix generation (LRMG) leads to a decrease in performance, as LRMG helps to transform edge prior knowledge into a more informative global space for the main network to utilize edge prior knowledge effectively.

3) Performance of PFDM: To assess the effectiveness of the PFDM, the ablation experiments are conducted as shown in Table III. The results show that the PFDM significantly outperforms traditional series connections (SCs) and parallel connections (PCs) due to its progressive distillation connection, which enhances feature refinement and extraction. Additionally, the SCU consisting of 1×3 and 3×1 convolutions is used to replace the 3×3 convolution. This design enables the network to capture a range of feature information in both horizontal and vertical directions. This separation also improves the network's ability to model directional features, thereby improving overall performance and feature extraction capability. Removal of the combined attention block (CAB) resulted in a decrease in performance, highlighting its essential

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TABLE IV

QUANTITATIVE COMPARISON OF THE QUICKBIRD DATASET ON SIMULATED DATA. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE

	Methods	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
SISR	Bicubic [61]	0.1011	11.5195	4.1318	0.1798	0.1024
	FeNet [40]	0.1353	11.2934	4.2925	0.1683	0.1369
	CTNet [62]	0.1782	10.7739	4.0563	0.1959	0.1763
	HSENet [63]	0.1849	10.2202	4.1383	0.1830	0.1389
	HAUNet [64]	0.2193	10.8497	4.0731	0.1580	0.2213
	TTST [43]	0.2451	10.7689	4.0606	0.1984	0.2377
Two-stage	AWLP [65]+TTST [43]	0.5293	10.6804	3.1695	0.4247	0.5371
	BRHM [66]+TTST [43]	0.5090	10.0431	3.1339	0.4373	0.5167
	FusNet [32]+TTST [43]	0.6661	8.9345	2.7380	0.5499	0.6641
	U2Net [67] +TTST [43]	0.6678	9.0981	2.7418	0.5441	0.6681
	S2DBPN [35]+TTST [43]	0.6456	9.3145	2.8215	0.5138	0.6463
	AWFLN [68]+TTST [43]	0.6700	8.9873	2.7487	0.5419	0.6699
End-to-end	Ours	**0.6952**	**8.6969**	**2.5728**	**0.6194**	**0.6955**

TABLE V

QUANTITATIVE COMPARISON OF THE GAOFEN-2 DATASET ON SIMULATED DATA. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE

	Methods	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
SISR	Bicubic [61]	0.3871	3.2204	1.0265	0.6504	0.3876
	FeNet [40]	0.4340	3.2166	1.0018	0.6539	0.4265
	CTNet [62]	0.4685	3.0812	1.0055	0.6517	0.4649
	HSENet [63]	0.4454	3.5659	1.1330	0.6491	0.3994
	HAUNet [64]	0.4670	3.1220	1.0107	0.6373	0.6373
	TTST [43]	0.4706	3.0931	1.0073	0.6707	0.6631
Two-stage	AWLP [65]+TTST [43]	0.6410	3.9822	1.0969	0.7040	0.6202
	BRHM [66]+TTST [43]	0.6784	3.7307	1.0000	0.7223	0.6609
	FusNet [32]+TTST [43]	0.7420	2.6983	0.7072	0.7942	0.7171
	U2Net [67]+TTST [43]	0.7841	2.6176	0.6343	0.7982	0.7813
	S2DBPN [35]+TTST [43]	0.7641	2.6174	0.6362	0.7944	0.7608
	AWFLN [68]+TTST [43]	0.7779	2.5823	0.6271	0.7987	0.7762
End-to-end	Ours	**0.8087**	**2.1812**	**0.5591**	**0.8318**	**0.8092**

TABLE VI

QUANTITATIVE COMPARISON OF THE WORLDVIEW-3 DATASET ON SIMULATED DATA. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE

	Methods	Q↑	SAM↓	ERGAS↓	SCC↑	Q_N↑
SISR	Bicubic [61]	0.2522	3.9337	1.9668	0.3716	0.2503
	FeNet [40]	0.2623	4.1447	6.8426	0.3696	0.1140
	CTNet [62]	0.2985	3.8226	6.6206	0.3817	0.1215
	HSENet [63]	0.2131	5.9680	7.7535	0.3675	0.0942
	HAUNet [64]	0.2560	4.0384	6.7337	0.3363	0.1081
	TTST [43]	0.2378	5.1405	8.0951	0.3715	0.0981
Two-stage	AWLP [65]+TTST [43]	0.5888	3.9966	1.6459	0.5687	0.5709
	BRHM [66]+TTST [43]	0.6060	3.5615	1.6528	0.5772	0.5916
	FusNet [32]+TTST [43]	0.6366	3.0119	1.2891	0.6399	0.6323
	U2Net [67]+TTST [43]	0.6404	3.0209	1.2882	0.6413	0.6384
	S2DBPN [35]+TTST [43]	0.6113	3.2089	1.3751	0.5770	0.6058
	AWFLN [68]+TTST [43]	0.6357	3.0573	1.3179	0.6182	0.6325
End-to-end	Ours	**0.6547**	**2.8592**	**1.2420**	**0.6675**	**0.6503**

role in enhancing features and preserving critical details. Removal of hierarchical feature enhancement (HFM), i.e., direct feature extraction via convolution, results in a significant drop in performance. This is because HFM captures hierarchical details that are essential to the performance of the model. Overall, the PFDM excels by integrating progressive distillation connections with SCUs and CAB, resulting in superior feature representation.

D. Experiments on Simulated Data

1) QuickBird Dataset: The average quantitative results of the 20 testing MSIs and PAN images for the QuickBird dataset are shown in Table IV, with bold and underline indicating the best and second best values. First, compared to the SISR methods, the proposed method shows superior performance by incorporating additional spatial information.

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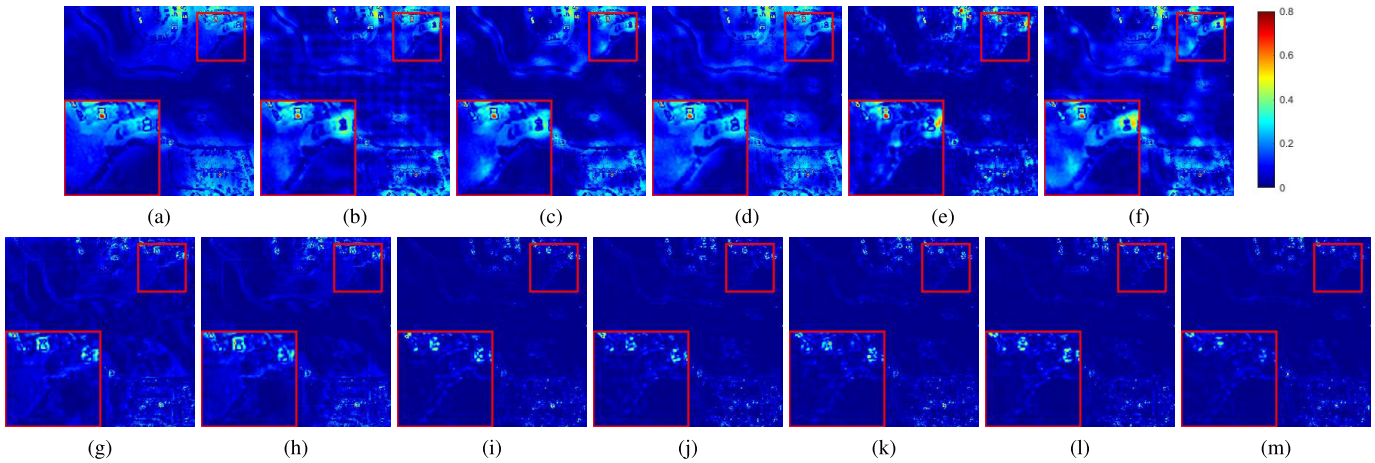


Fig. 5. Absolute error maps of different methods on the QuickBird dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

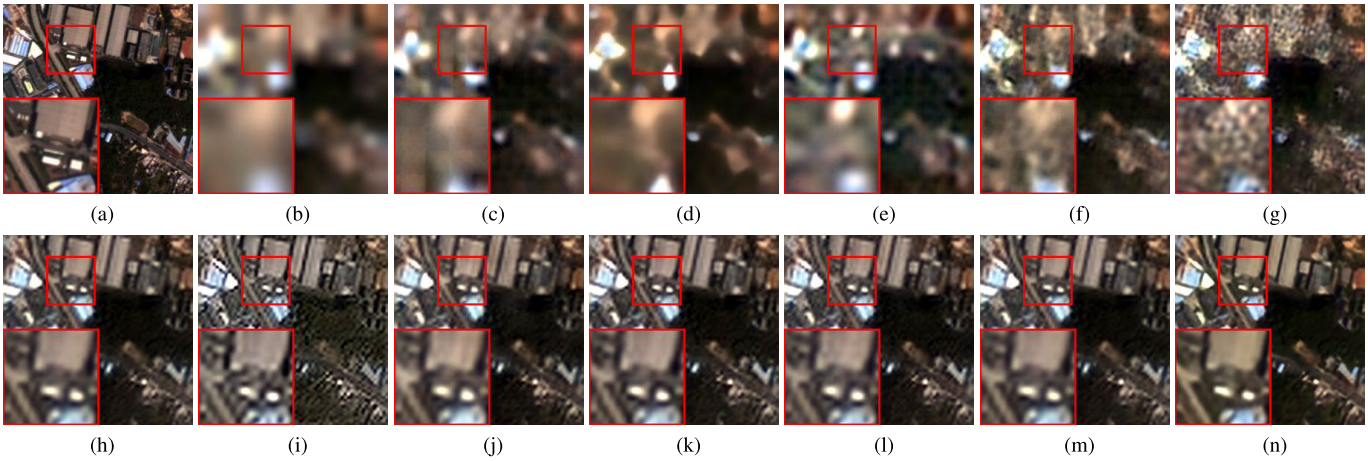


Fig. 6. Reconstruction results of simulation experiments on the GaoFen-2 dataset. (a) GT. (b) Bicubic. (c) FeNet. (d) CTNet. (e) HSENet. (f) HAUNet. (g) TTST. (h) AWLP + TTST. (i) BRHM + TTST. (j) FusNet + TTST. (k) U2Net + TTST. (l) S2DBPN + TTST. (m) AWFLN + TTST. (n) Ours.

As depicted in Table IV, FSRNet outperforms SISR methods across all five metrics. Particularly, it shows an improvement of 0.4501/2.072/1.4878/0.4210/0.4578 in terms of Q/SAM/ERGAS/SCC/Q_N as compared to TTST. This is attributed to the fact that traditional SISR methods do not effectively utilize the spatial information from PAN images, leading to challenges in achieving good results at a large super-resolution scale (e.g., 16). Moreover, a comparison with a two-stage method combining the best RSIF and SISR methods still validates the superior performance of FSRNet across all metrics, showing improvements of 0.0252/0.2904/0.1759/0.0775/0.0256 in Q/SAM/ERGAS/SCC/Q_N compared to AWFLN + TTST. This discrepancy may be due to the cumulative errors in directly cascaded two-stage methods, resulting in inferior performance compared to the proposed end-to-end FSRNet.

The qualitative evaluation of the QuickBird dataset is depicted in Figs. 4 and 5, showcasing the reconstructed images and absolute error maps. The absolute error maps are colored blue, with bluer regions corresponding to better reconstructed images. Fig. 4 illustrates that FSRNet generates more accurate contours compared to other comparison methods. This may be due to the fact that the proposed IEC scheme can efficiently calibrate and sharpen the structure of MSIs using gradient prior knowledge. Furthermore, in Fig. 5, the absolute error map of FSRNet exhibits smaller errors, indicating a closer proximity to the GT. This qualitative evaluation aligns with the quantitative results, underscoring the superior performance of FSRNet over previous methods.

2) GaoFen-2 Dataset: The quantitative and qualitative results for the GaoFen-2 dataset are detailed in Table V and in Figs. 6 and 7. The results of multiple SISR methods (i.e., FeNet, CTNet, HSENet, HAUNet, and TTST) are unsatisfactory. This is due to the difficulty of directly learning the mapping relationship between LR to HR images at large scale factors. As illustrated in Table V, the proposed method outperforms SISR and two-stage methods. Specifically, the proposed method shows improvements compared to the best SISR method (i.e., TTST) and the top two-stage method (i.e., AWFLN + TTST) in all metrics. These results highlight the benefits of the proposed end-to-end framework in reconstructing HR MSIs.

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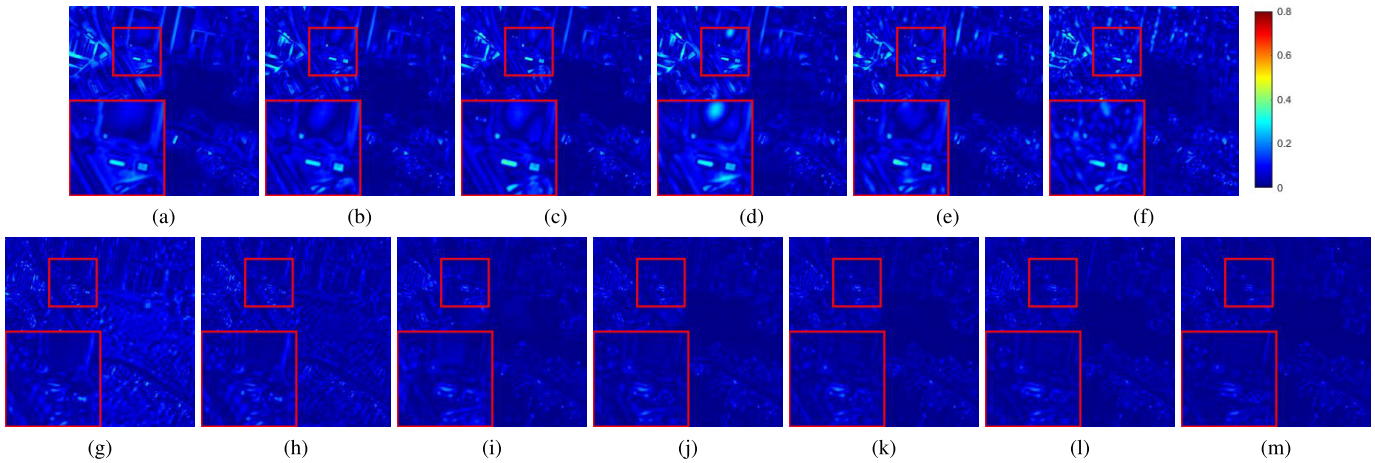


Fig. 7. Absolute error maps of different methods on the GaoFen-2 dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

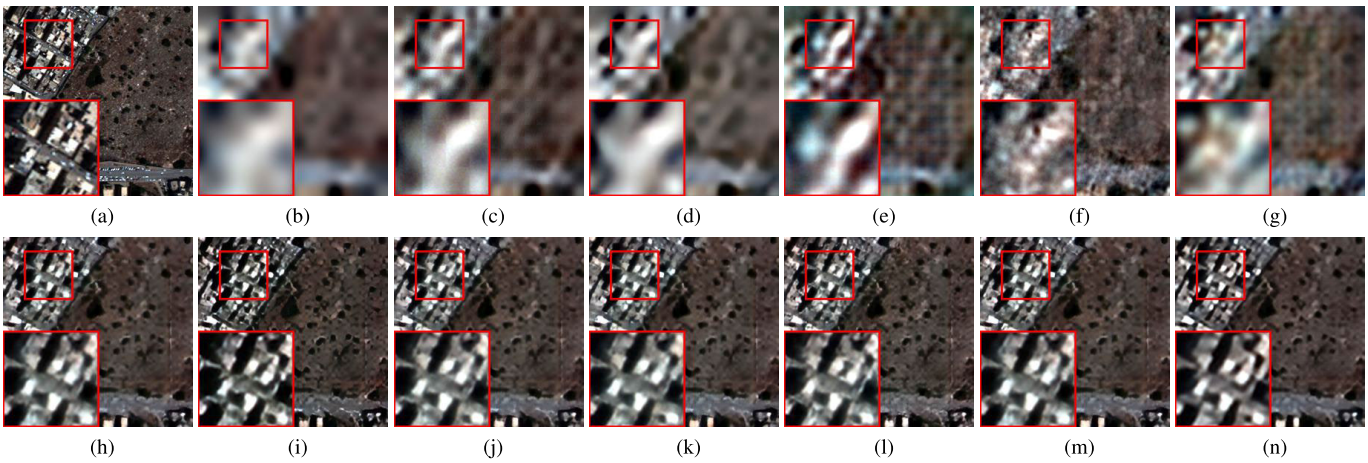


Fig. 8. Reconstruction results of simulation experiments on the WorldView-3 dataset. (a) GT. (b) Bicubic. (c) FeNet. (d) CTNet. (e) HSENet. (f) HAUNet. (g) TTST. (h) AWLP + TTST. (i) BRHM + TTST. (j) FusNet + TTST. (k) U2Net + TTST. (l) S2DBPN + TTST. (m) AWFLN + TTST. (n) Ours.

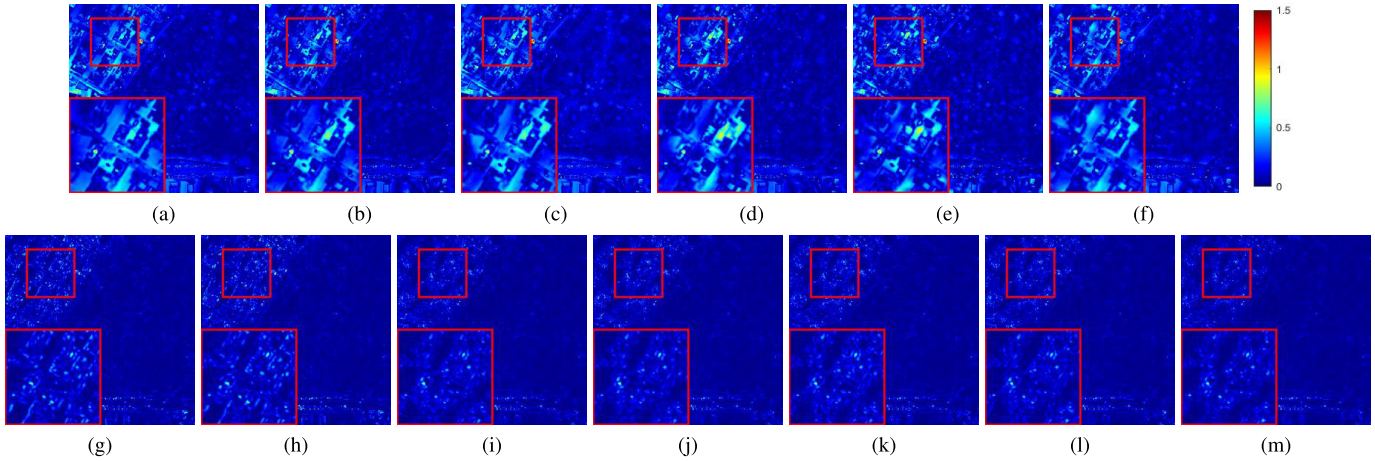


Fig. 9. Absolute error maps of different methods on the WorldView-3 dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

As shown in Fig. 6, from the enlarged red rectangular region, it can be observed that the results of FSRNet possess more accurate and clearer edge contours compared to the images reconstructed by other methods. The two-stage methods tend to produce more reconstruction errors, e.g., Fig. 6(i) and (j). This is because the RSIF process may introduce some noise or artifacts that may be amplified during the subsequent super-resolution reconstruction, leading to a decline in the image quality. As depicted in Fig. 7, it is evident from the enlarged red rectangular region that the results of FSRNet

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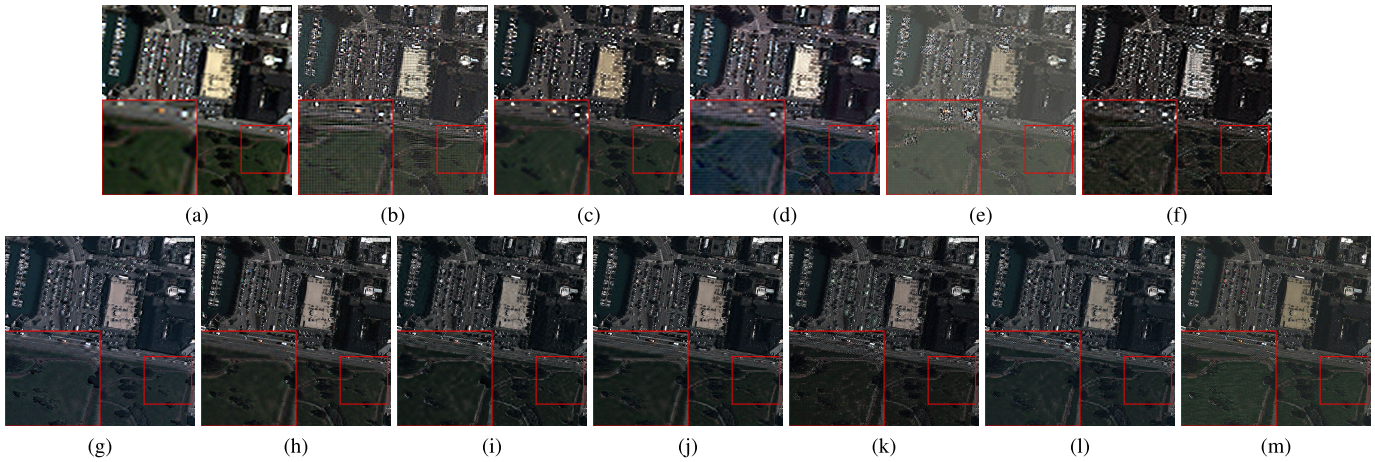


Fig. 10. Reconstruction results of real experiments on the QuickBird dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

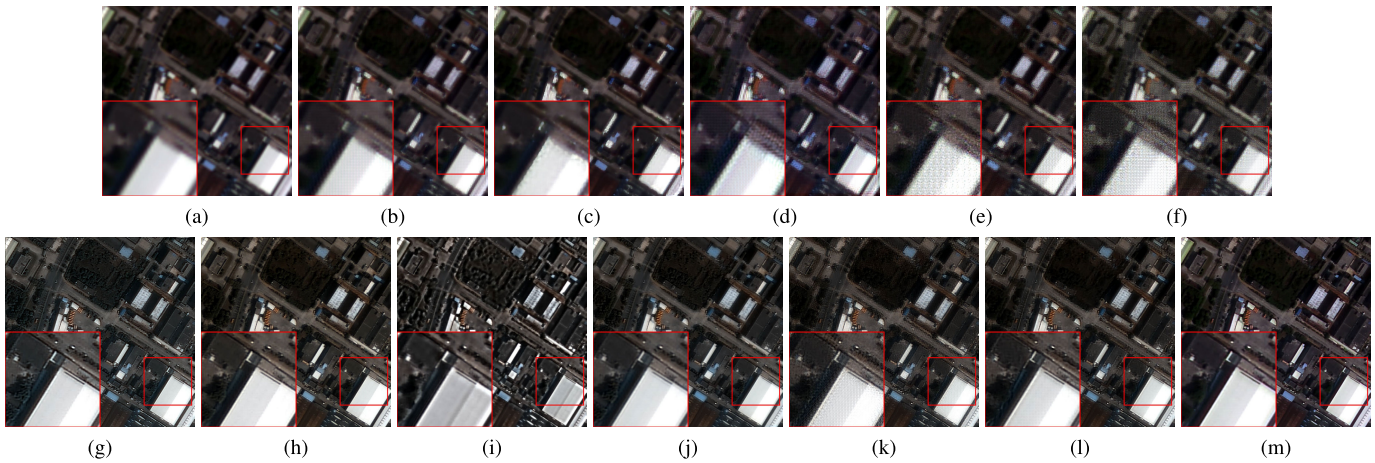


Fig. 11. Reconstruction results of real experiments on the GaoFen-2 dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

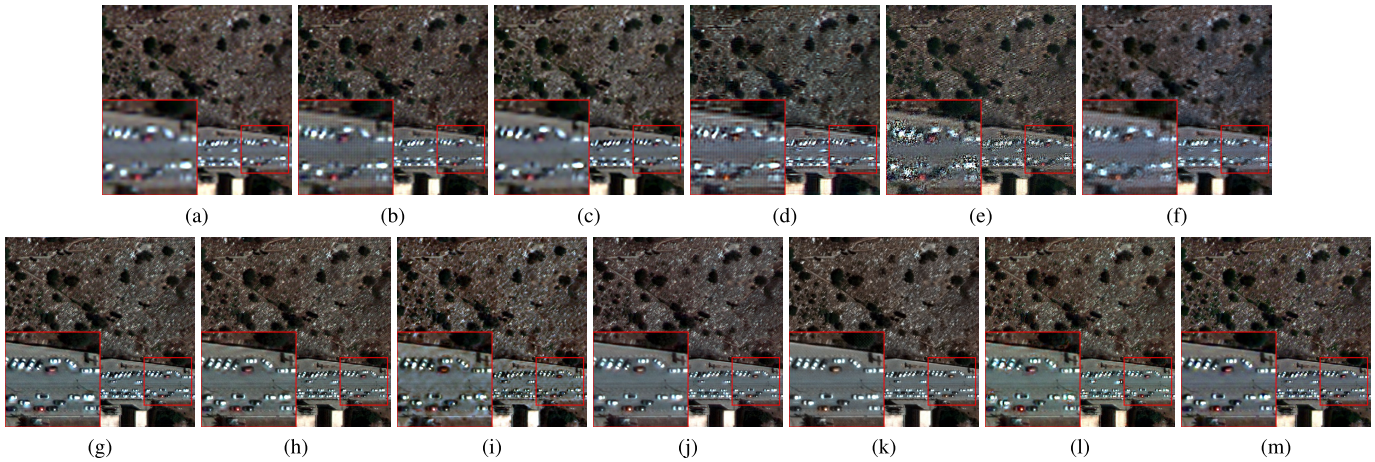


Fig. 12. Reconstruction results of real experiments on the WorldView-3 dataset. (a) Bicubic. (b) FeNet. (c) CTNet. (d) HSENet. (e) HAUNet. (f) TTST. (g) AWLP + TTST. (h) BRHM + TTST. (i) FusNet + TTST. (j) U2Net + TTST. (k) S2DBPN + TTST. (l) AWFLN + TTST. (m) Ours.

exhibit smaller errors compared to other results. Therefore, the FSRNet is satisfactory in reconstructing HR MSI.

3) WorldView-3 Dataset: To further investigate the performance of FSRNet, experiments are conducted on the WorldView-3 dataset with spectral band of 8. As presented in Table VI, FSRNet achieves optimal quantitative results across all image quality metrics. Specifically, it outperforms the second best two-stage method, U2Net + TTST, by significant margins of 0.0143 in Q, 0.1617 in SAM, 0.0462 in ERGAS, 0.0262 in SCC, and 0.0119 in Q_N. This superiority not

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TABLE VII
QUANTITATIVE COMPARISON OF THE THREE DATASETS ON REAL DATA. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE

Datasets	QuickBird	GaoFen-2	WorldView-3
Methods	NIQE↓/PIQE↓/D_λ↓/D_s↓/QNR↑	NIQE↓/PIQE↓/D_λ↓/D_s↓/QNR↑	NIQE↓/PIQE↓/D_λ↓/D_s↓/QNR↑
Bicubic [61]	7.1226/92.1034/-/0.1878/-	7.6762/95.4201/-/0.2630/-	7.1794/97.3291/-/0.1930/-
FeNet [40]	7.5886/33.7399/0.1833/0.2659/0.5995	7.0150/49.1756/0.0319/0.2562/0.7201	8.6885/29.8654/0.2234/0.2464/0.5852
CTNet [62]	5.8142/88.4163/0.1061/0.2144/0.7022	6.1784/94.0818/**0.0280**/0.2452/0.7337	6.4761/93.0588/0.1841/0.2290/0.6291
HSENet [63]	5.9825/33.1898/0.1053/0.2160/0.7014	6.0743/51.8531/0.0348/0.2654/0.7090	6.5330/38.2939/0.2403/0.3080/0.5257
HAUNet [64]	7.4588/36.3449/0.1651/0.2621/0.6161	5.2276/49.5900/0.0674/0.3002/0.6526	5.1531/25.9813/0.2106/0.2605/0.5838
TTST [43]	6.6134/78.5065/0.1151/0.2284/0.6828	5.9619/50.7152/0.0738/0.3125/0.6368	6.7987/56.1601/0.1801/0.2100/0.6477
AWLP [65]+TTST [43]	4.8542/41.2960/0.1506/0.1584/0.7149	6.5352/49.2666/0.1458/0.1934/0.6890	4.7819/58.2457/0.1017/0.0991/0.8093
BRHM [66]+TTST [43]	4.9609/39.3656/0.1465/0.1638/0.7137	6.3584/48.4235/0.1585/0.2129/0.6623	4.5391/58.2317/0.1288/0.1354/0.7532
FusNet [32]+TTST [43]	6.2746/44.9315/0.1420/0.0371/0.8262	5.9837/42.4415/0.2120/0.1715/0.6529	4.8549/53.3949/0.1079/**0.0928**/0.8093
U2Net [67]+TTST [43]	4.9252/45.4349/0.1427/0.1696/0.7119	6.2176/55.3065/0.1503/0.2024/0.6777	4.8413/50.1930/0.1234/0.1627/0.7340
S2DBPN [35]+TTST [43]	7.0830/39.9544/0.1508/0.0699/0.7898	6.0768/59.2070/0.0826/0.2542/0.6842	4.6408/50.0681/0.1164/0.1573/0.7446
AWFLN [68]+TTST [43]	6.1369/44.2700/0.1246/0.0331/0.8464	6.3024/49.2026/0.0822/0.1885/0.7448	4.6068/55.0229/0.0699/0.0999/**0.8372**
Ours	**4.7974/29.5108/0.1049/0.0296/0.8686**	**4.4607**/47.8785/0.0643/**0.1679/0.7786**	**4.5265/21.5848/0.0691**/0.1442/0.7967

only highlights the effectiveness of the proposed end-to-end framework but also underscores its ability to avoid cumulative errors that are inherent in two-stage methods, thereby leading to superior reconstruction results.

The qualitative results of all the methods on WorldView-3 are illustrated in Figs. 8 and 9. As shown in the enlarged image in the red rectangular region of Fig. 8, FSRNet reconstructs relatively sharper details compared to other methods, while also more closely resembling the contour structure of GTs. While the absolute error images further demonstrate the advantages of FSRNet, as shown in Fig. 9, it can be seen that the result of FSRNet is the bluest, indicating that FSRNet is closest to GT. The reconstruction results of the proposed method on 8-band MSI remain the best, demonstrating the broad applications of FSRNet. These comprehensive quantitative and qualitative analyses demonstrate the robustness and wide applicability of FSRNet, especially in reconstructing 8-band MSIs.

E. Experiments on Real Data

In this section, all methods are applied to MSI reconstruction using real data to evaluate performance. The testing input images (i.e., MSIs and PAN images) are used without any degradation operation. For the QuickBird, GaoFen-2, and WorldView-3 datasets, the networks trained in the corresponding simulation experiments are, respectively,deployed for the experiments on real data.

Fig. 10 presents the reconstructed HR MSI on the real QuickBird dataset. The reconstructed MSIs by SISR methods, such as FeNet, CTNet, HSENet, HAUNet, and TTST exhibit smooth and blurred spatial details. Two-stage methods like AWFLN + TTST introduce spectral distortion. In contrast, the proposed FSRNet exhibits superior spatial detail while preserving better spectral fidelity. Fig. 11 displays the reconstructed MSI of the GaoFen-2 dataset using real data. The SISR methods exhibit poor reconstruction performance due to the large scale factor and the absence of HR auxiliary image guidance. The two-stage methods also display noise points due to cumulative errors, impacting overall image quality. As depicted in the enlarged region, the proposed method achieves favorable visual effects, capturing accurate and natural texture details. The reconstruction results on the WorldView-3 dataset are presented in Fig. 12. The SISR methods struggle with spatial reconstruction, resulting in low image quality. FSRNet performs excellently in retaining spatial details, and the contours of the cars in the figure are more complete.

TABLE VIII
COMPUTATIONAL COST FOR ALL METHODS ON THE QUICKBIRD DATASET

Methods	Params. (M)	FLOPs (G)	Time (s)
Bicubic [61]	-	-	0.010
FeNet [40]	0.789	0.197	0.019
CTNet [62]	0.360	0.903	0.022
HSENet [63]	5.729	4.364	0.026
HAUNet [64]	10.571	1.620	0.049
TTST [43]	18.602	7.956	0.103
AWLP [65]+TTST [43]	- +18.310	- +76.890	0.024+0.101
BRHM [66]+TTST [43]	- +18.310	- +76.890	0.013+0.101
FusNet [32]+TTST [43]	0.150+18.310	1.210+76.890	0.003+0.101
U2Net [67] +TTST [43]	0.210+18.310	2.450+76.890	0.005+0.101
S2DBPN [35]+TTST [43]	16.188+18.31	10.660+76.89	0.008+0.101
AWFLN [68]+TTST [43]	0.081+18.310	0.321+76.890	0.006+0.101
Ours	1.130	13.810	0.010

In addition, the quantitative metrics from real experiments on the three datasets are shown in Table VII, including the NIQE, PIQE, D_λ, D_s, and QNR value. The proposed FSRNet has the best values in terms of five metrics on the QuickBird dataset. Typically, it shows an improvement of 0.0568/3.679/0.0004/0.0035/0.0222 in terms of NIQE/PIQE/D_λ/D_s/QNR as compared to the second-best values. Meanwhile, it can be seen that the proposed method achieves the best results in most of the five metrics in GaoFen-2 and WorldView-3 datasets. This is consistent with the qualitative results of experiments on real data, which further proves the advantages of FSRNet in experiments on real data.

F. Computational Cost

To analyze the computational cost of all the compared methods, the number of parameters (Params.), floating point operations per second (FLOPs), and inference time for all methods are given in Table VIII. Specifically, the three

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metrics for all methods in the QuickBird dataset are obtained. As shown in Table VIII, the proposed end-to-end method outperforms the two-stage method (i.e., U2Net + TTST, S2DBPN + TTST, and AWFLN + TTST) in all three metrics regarding computational cost. This is because the proposed method integrated optimization of tasks, eliminating the need for separate stages and reducing redundant computations. Compared to the SISR method, the proposed method can obtain better image quality at essentially the same computational cost.

V. Conclusion

In conclusion, this study proposes a pioneering end-to-end framework, FSRNet, specifically designed for MSI reconstruction. It successfully overcomes the limitations of traditional RSIF and SISR methods. By seamlessly integrating RSIF and SISR into a unified framework, FSRNet can reconstruct higher-resolution MSI, mitigating the cumulative errors that often arise in the two-stage cascaded methods. Then, an innovative IEC scheme is proposed to implicitly calibrate the edge structures of MSIs in the gradient domain, ensuring that reconstructed images are rich in detail and high in accuracy. The IEC scheme effectively calibrates the edge structural features of the main network during training and is omitted during testing, thus not increasing network complexity in practice. In addition, PFDM progressively extracts and refines features at multiple levels, facilitating the capture of complex structures within MSI. Extensive experiments and comparisons show that the proposed FSRNet consistently outperforms existing methods on the QuickBird, Gaofen-2, and WorldView-3 datasets. In particular, the effectiveness of each component in FSRNet is also validated by ablation experiments.

Despite the significant contributions of FSRNet to the improvement of the image quality, this study primarily focuses on the low-level tasks, without exploring the relationship between MSI reconstruction and downstream high-level tasks such as classification and change detection. In the future, to further enhance the practical application value of MSI reconstruction, we are committed to integrating MSI reconstruction with downstream tasks and fully exploring the intrinsic relationship between high-level and low-level tasks.

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